

Ewan Burnett

PROGRAMMER

CONTACT

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ABOUT ME

I'm a dedicated Computer Scientist, with a love for learning. Passionate about delivering efficient, well-engineered software for use in both Real-time and offline simulation. As a Reliable developer, I'm looking to contribute my skills and creativity in professional development within the Computer Graphics industry.

LANGUAGE PROFICIENCY

- | | |
|----------------|--------|
| ▪ C | ▪ Lua |
| ▪ C++ | ▪ BASH |
| ▪ Rust | ▪ HLSL |
| ▪ C# | ▪ GLSL |
| ▪ Python | ▪ ISPC |
| ▪ X86 Assembly | |

API KNOWLEDGE

- Vulkan
- DirectX 12
- DirectX 11
- OpenGL
- PS5 API
- .NET

SOFTWARE COMPETENCY

- Unreal Engine 5
- Unity Engine
- Visual Studio
- Jira
- Microsoft Office Suite
- Renderdoc
- Github
- Autodesk Maya
- NVIDIA Nsight

COMMUNICATION

- English (Native)
- Japanese (JLPT N4)
- German (GCSE)

PROJECTS

MASTERS' DISSERTATION – IMPLICIT SURFACE RENDERING USING MESH SHADERS

September 2025 – December 2025

- Research into the efficacy of Real-Time Triangulation of implicit surfaces using Turing Mesh Shaders.
- Utilizes Dual-Contouring to triangulate surfaces in a massively parallel context, taking full advantage of Modern Graphics Hardware.
- Implemented using Vulkan 1.3 and C++ 20

Lapis - PS5 Game Engine + Group Project

September 2024 – May 2025

- Collaborated in a team of Programmers, Artists and Designers to develop a Game Engine and Game project from scratch, targeting the PlayStation 5.
- Worked with Proprietary PlayStation 5 Development kits, SDKs and software, integrating toolchains for compilation and package installation.
- Implemented an Immediate-mode renderer, multi-threaded Audio manager, Lua Scripting backend, and A* Pathfinding for use in the Game project.

UNDERGRADUATE DISSERTATION – CASCADED REAL TIME GLOBAL ILLUMINATION

September 2024 – February 2025

- Implemented in Vulkan with C++ 17, for Windows, Linux and Android.
- Exploratory research into the viability of simulating Global Illumination without Hardware accelerated Ray Tracing, for use on lower-end hardware.
- Utilises Spherical harmonics-based Irradiance Probes to encode the environment's luminance in a hardware efficient manner.

PROFESSIONAL EXPERIENCE

Student Mentor

Sheffield Hallam University

September 2022 – May 2026

Part-time role as a part of the Mentoring team in the Department of Computing.

- Providing support and positive feedback for first, second and third-year mentees.
- Hosting weekly workshops and 1-1 sessions to reinforce knowledge of advanced topics.
- Collaborating with Faculty and the Mentoring team to host competitions and events.
- Adapting my existing knowledge to aid students in different fields of computing.

Student Ambassador

Sheffield Hallam University

September 2021 – May 2022

Part-time role as a Student Ambassador for the Department of Computing.

- Running Open Days within a large team of ambassadors.
- Preparing the course booth, and tech demos of coursework for prospective students.
- Hosting presentations about the available courses at the university for the public.
- Engaging with prospective students with the goal of promoting the university and its courses.

References Available on Request

KEY SKILLS

Computer Architecture

Strong knowledge of how Computer Hardware works; From Assembly Language to GPU Architecture.

Parallel Software Development

Adept at the development of parallelised software using SIMD, Multi-threading and GPU Compute.

Graphics Programming

Familiarity with Rasterization and Ray Tracing Pipelines for use in Real-time Rendering, alongside the theory driving modern Computer Graphics.

Cross Platform Development

Practical experience in developing Cross-platform Software for Windows, Linux and Sony's PlayStation 5™.

Learned and Adaptable

I'm driven by a strong desire to learn; and am subsequently able to quickly familiarise myself with new development environments and codebases.

Game Engine Architecture

Proficient in the development of the underlying subsystems driving modern Game Engines; From Simulation to Graphics.

Agile Development

Successfully developed software in Agile teams, using software such as Trello to delegate and manage tasks within sprints.

Version Control

Proficient in the use of Git for Software Version Control and organisation.

Debugging

Able to effectively debug software using Industry-standard tools and techniques.

EDUCATION

MComp Computer Science for Games | 1st

Sheffield Hallam University

September 2020 – May 2026

Integrated Master's specialising in Software Engineering and Game Engine development.

- Advanced C++ Programming, focused on Real-Time Rendering and Simulation.
- Linear Algebra and Calculus for 3D Graphics
- Opportunity to publish proprietary games software on the PlayStation 5.
- Collaborative work with Artists and Designers to create games software from scratch.

BTEC Games Technology | Distinction

Confetti Institute of Creative Technologies

September 2018 – June 2020

Creative Digital Media Level 3 diploma focused on Game Production.

- Studied Foundational Game Design and Development Principles
- Scripted Dynamic behaviour with C# in Unity and C++ in Unreal Engine 4
- Explored the 3D Asset creation pipeline with Autodesk Maya